



ISLAMIC UNIVERSITY OF TECHNOLOGY

Computer Science and Engineering (CSE)

Chem 4241: Chemistry



Chemical Bonding

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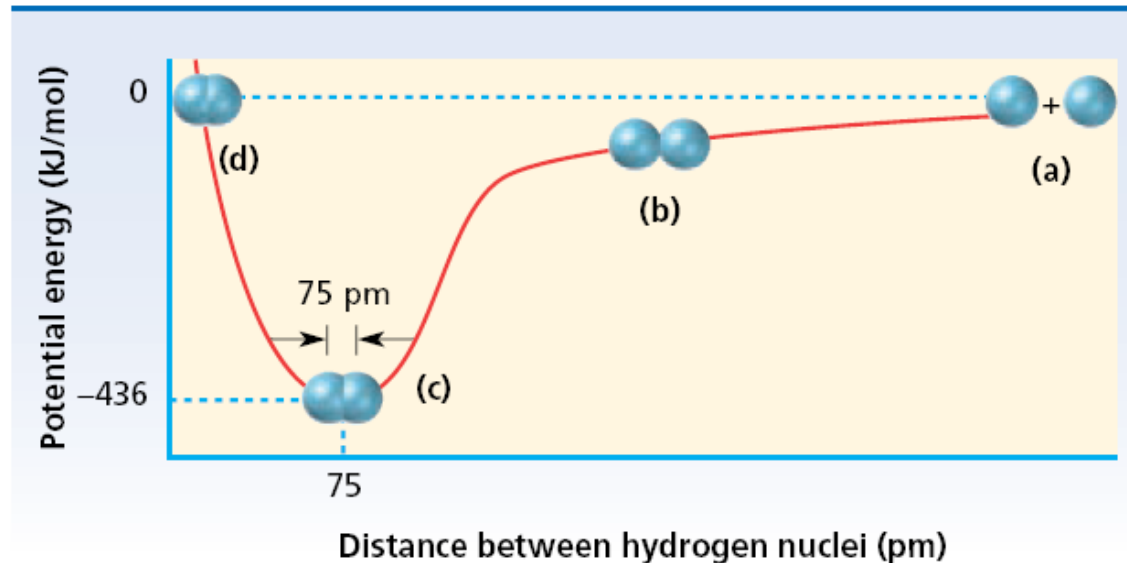
Chemical Bond



A *chemical bond* is a strong attractive force that exists between certain atoms in a substance.

Why do atoms form chemical bonds???

- As independent particles, atoms are at relatively high potential energy, less stable.
- Nature always favors the arrangements in which potential energy is minimized, more stable.
- By bonding with each other, the potential energy decreased, thereby creating more stable arrangements of matter.





Types of Chemical Bond



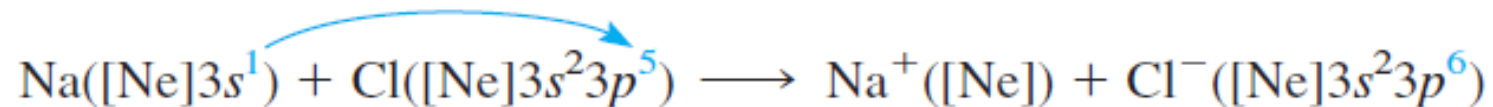
There are three different types of bonds recognized by chemists :

1. Ionic bond
2. Covalent bond
3. Coordinate covalent bond

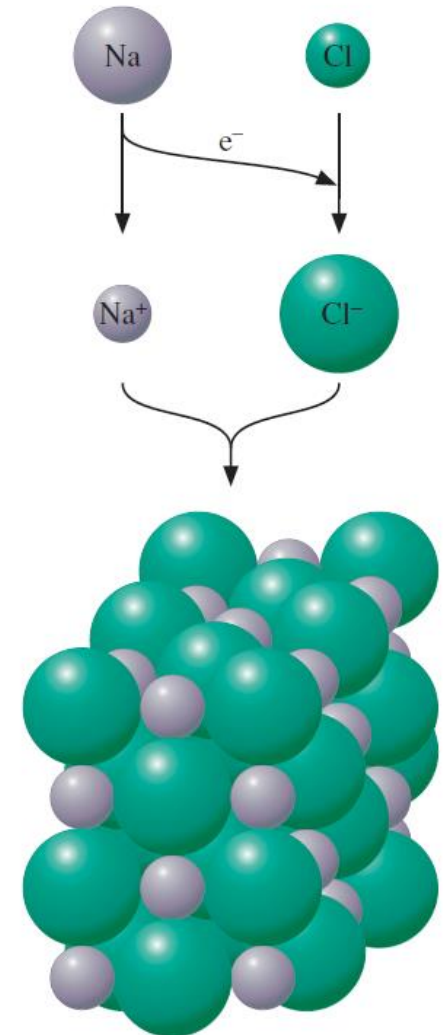
There is fourth type of bond, called metallic bond, we will discuss later.

Ionic bond

An *ionic bond* is a chemical bond formed by the electrostatic attraction between positive and negative ions. The bond forms between two atoms when one or more electrons are transferred from the valence shell of one atom to the valence shell of the other. The atom that loses electrons becomes a cation (positive ion), and the atom that gains electrons becomes an anion (negative ion).



Within the sodium chloride crystal, NaCl, every Na^+ ion is surrounded by six Cl^- ions, and every Cl^- ion by six Na^+ ions.

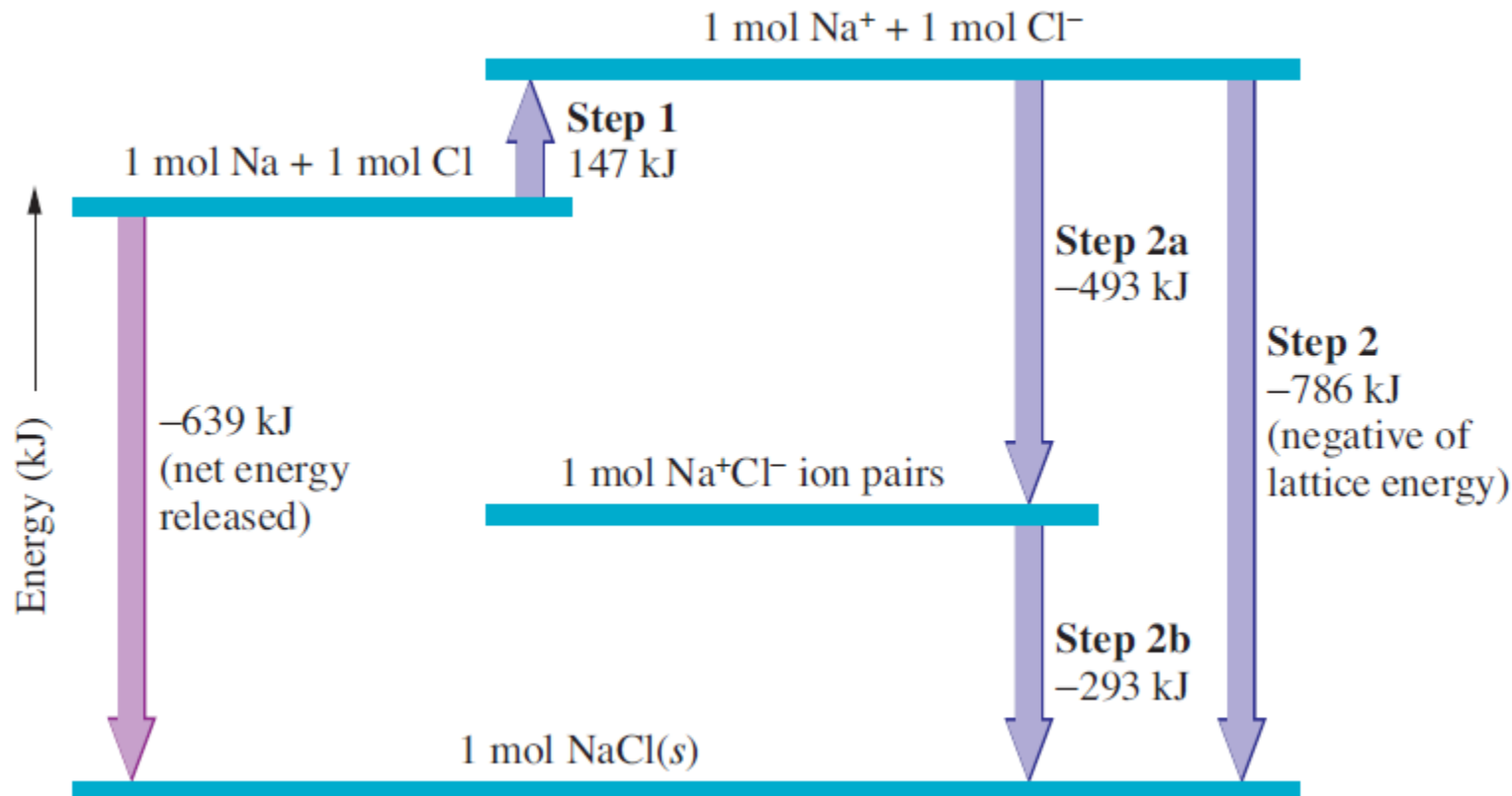
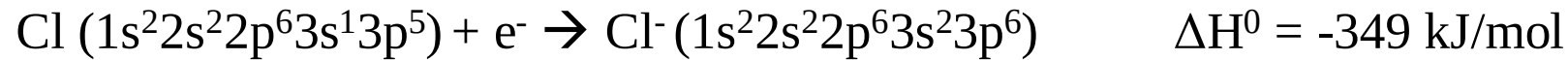
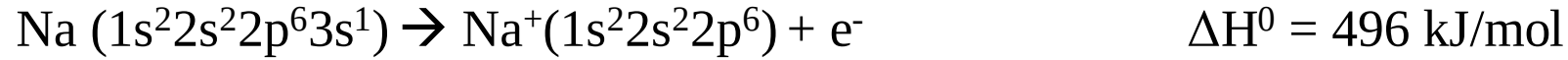




Energy Involved in Ionic Bonding



If atoms come together and bond, there should be a net decrease in energy, because the bonded state should be more stable and therefore at a lower energy level.



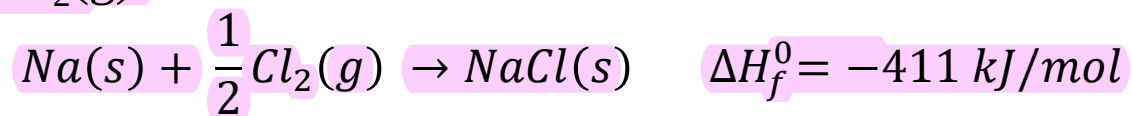
The *lattice energy* is the change in energy that occurs when an ionic solid is separated into isolated ions in the gas phase. For sodium chloride, the process is $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$



Lattice Energies from the Born–Haber Cycle

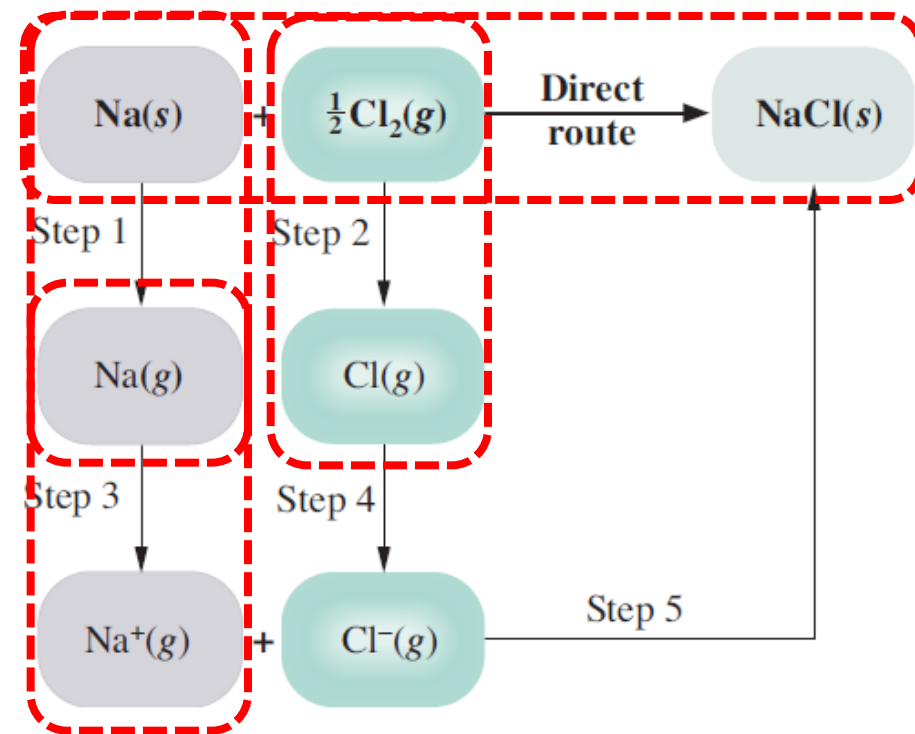


To obtain the lattice energy of NaCl, let the solid sodium chloride being formed from the elements by two different routes, as shown in Figure. In one route, NaCl(s) is formed directly from the elements, Na(s) and $\frac{1}{2}\text{Cl}_2(\text{g})$.



The second route consists of five steps:

1. *Sublimation of sodium.* Metallic sodium is vaporized to a gas of sodium atoms. (*Sublimation* is the transformation of a solid to a gas.) The enthalpy change for this process, measured experimentally, is 108 kJ per mole of sodium.
2. *Dissociation of chlorine.* Chlorine molecules are dissociated to atoms. The enthalpy change for this equals the Cl-Cl bond dissociation energy, which is 240 kJ per mole of bonds, or 120 kJ per mole of Cl atoms.
3. *Ionization of sodium.* Sodium atoms are ionized to Na^+ ions. The enthalpy change is essentially the ionization energy of atomic sodium, which equals 496 kJ per mole of Na.

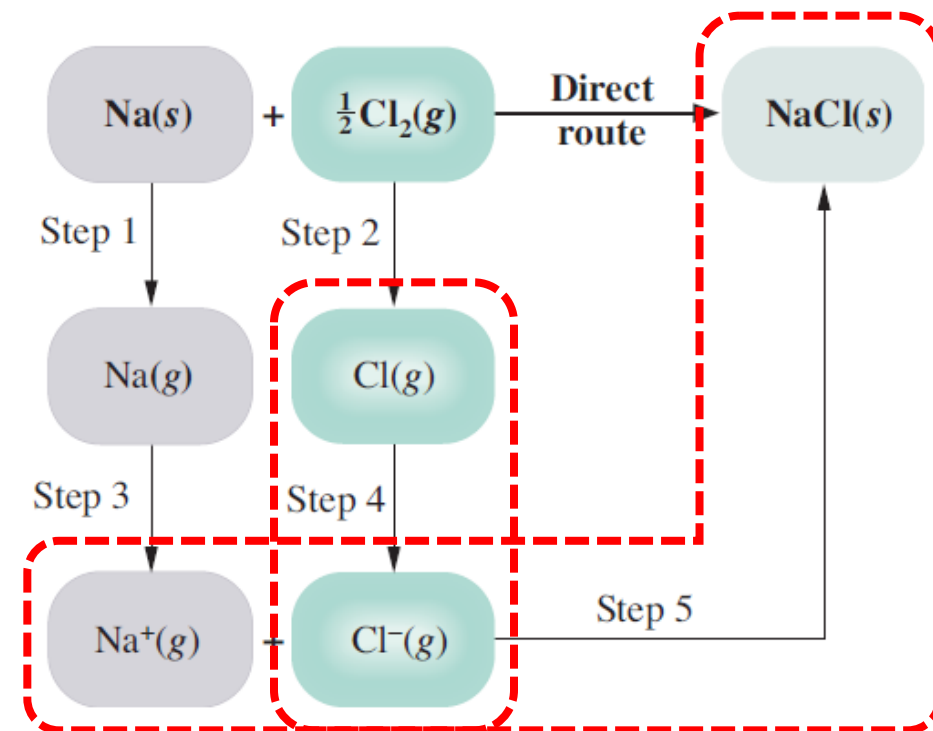




Lattice Energies from the Born–Haber Cycle



4. *Formation of chloride ion.* The electrons from the ionization of sodium atoms are transferred to chlorine atoms. The enthalpy change for this is the electron affinity of atomic chlorine, which equals -349 kJ per mole of Cl atoms.
5. *Formation of NaCl(s) from ions.* The ions Na^+ and Cl^- formed in Steps 3 and 4 combine to give solid sodium chloride. Because this process is just the reverse of the one corresponding to the lattice energy (breaking the solid into ions), the enthalpy change is the negative of the lattice energy. If we let U be the lattice energy, the enthalpy change for Step 5 is $-U$.

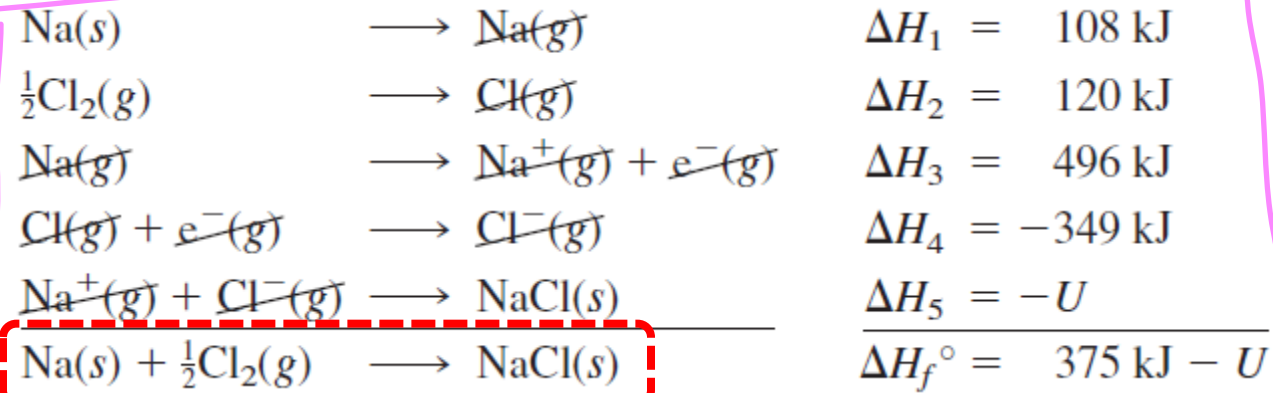




Lattice Energies from the Born–Haber Cycle



Let's write these five steps and add them, and also add the corresponding enthalpy changes, following Hess's law.



The final equation is simply the formation reaction for NaCl(s). The enthalpy of formation has been determined calorimetrically and equals -411 kJ.

Equating these two values, we get

$$375 \text{ kJ} - U = -411 \text{ kJ}$$

Solving for U yields the lattice energy of NaCl:

$$U = (375 + 411) \text{ kJ} = 786 \text{ kJ}$$



Factors Favoring The Formation of Ionic Bonds

1. **Number of valence electrons:** The atom forming the cation should possess 1, 2 or 3 valence electrons, while the other atom forming the anion should have 5, 6 or 7 valence electrons. The elements of group IA, IIA and IIIA satisfy this condition for cation and those of groups VA, VIA, and VIIA satisfy this condition for anion.
2. **Net lowering of Energy:** To form a stable ionic compound, there must be a net lowering of energy. In other words, energy must be released as a result of the electron transfer and formation of ionic compound by the following steps :
 - a) The removal of electron from an atom requires energy, which is described by ionization energy (IE). Lower the ionization energy greater will be the tendency of the metal atom to form cation and hence greater will be the ease of formation of ionic bond.
 - b) The addition of electron to an atom releases energy, which is described by electron affinity (EA). The higher the electron affinity, the more energy is released, resulting in a more stable anion formation which favors the formation of ionic bond.
 - c) The electrostatic attraction between cations and anions in the solid compound releases energy, which is related to the lattice energy. Greater lattice energy corresponds to a stronger ionic bond.



Factors Favoring The Formation of Ionic Bonds



The value of lattice energy depends on two main factors:

- **Size of the ions:** The force of attraction between cations and anions is inversely proportional to the square of the distance between them. Therefore, smaller ion sizes result in greater attraction and higher lattice energy.
- **Charge on ions:** The strength of the ionic bond is directly proportional to the charge on the ions. Greater charges on the ions lead to stronger attraction and higher lattice energy.

If the energy released in steps (b) and (c) is greater than the energy consumed in step (a), the overall process of electron transfer and formation of the ionic compound results in a net release of energy, contributing to the stability of the ionic bond.

3. **Electronegativity difference of A and B**

The formation of an ionic bond typically requires a significant difference in electronegativity between the atoms involved, generally greater than 2. This large difference ensures that one atom strongly attracts electrons, becoming a negatively charged ion (anion), while the other atom loses electrons to become a positively charged ion (cation). For example, sodium (Na) has an electronegativity of 0.9, while chlorine (Cl) has an electronegativity of 3.0. The difference in electronegativity between Na and Cl ($3.0 - 0.9 = 2.1$) is sufficient for them to form an ionic bond.



Characteristics of Ionic Compounds

➤ Solids at Room Temperature

Due to the strong electrostatic attraction between oppositely charged ions, they are held firmly in place within the crystal lattice. This strong bond prevents them from moving freely, a characteristic of the liquid state, thereby causing the substance to exist as a solid at room temperature.

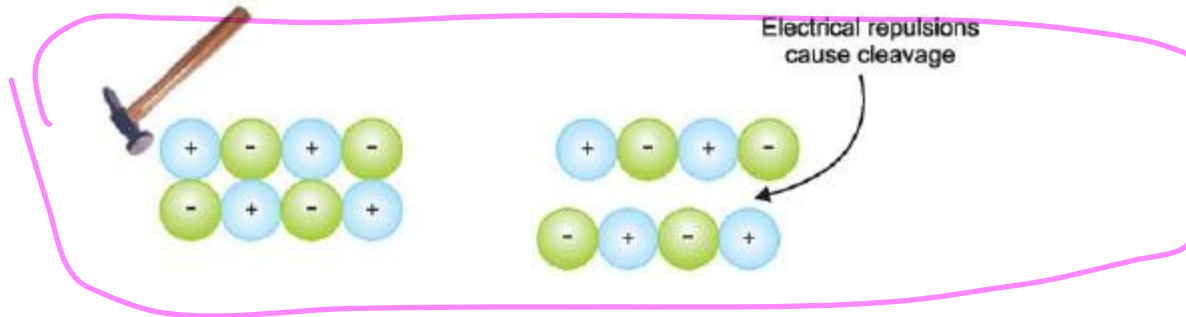
➤ High Melting Points

Ionic compounds typically exhibit high melting and boiling points. This is due to the strong electrostatic forces of attraction between the positively and negatively charged ions in the lattice structure. At high temperatures, the ions gain enough kinetic energy to overcome these attractive forces, allowing them to break free from their fixed positions in the lattice and move more freely, as seen in a liquid state.

➤ Hard and brittle

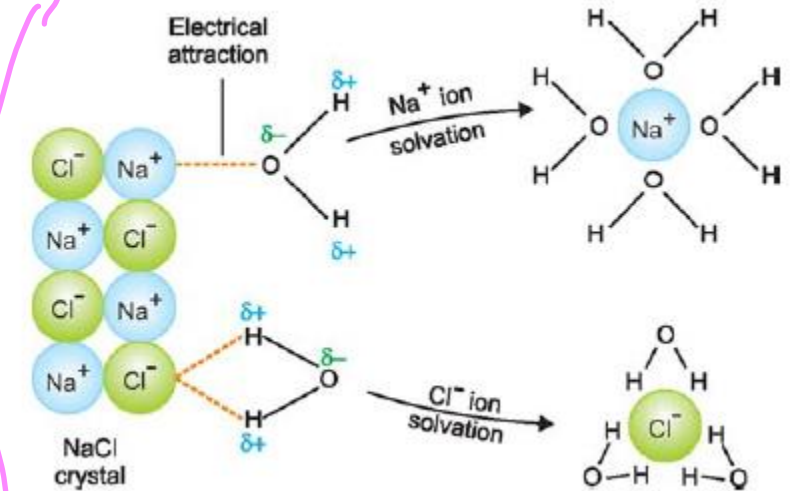
The crystals of ionic substances are hard and brittle due to the strong electrostatic forces that hold each ion in its designated position within the crystal lattice. These crystals are composed of layers of positively charged ions and negatively charged ions arranged alternately so that ions of opposite charges in adjacent layers align. When an external force is applied to a layer of ions relative to the next layer, even a slight shift can bring like-charged ions into proximity. This proximity leads to repulsion between the similarly charged ions, causing the crystal to fracture along planes of weakness. This phenomenon is known as **crystal cleavage**.

Characteristics of Ionic Compounds



➤ Soluble in water

When a crystal of an ionic substance is immersed in water, the polar water molecules exert an electrostatic pull that detaches the positively charged (+) and negatively charged (–) ions from the crystal lattice. These liberated ions become surrounded by water molecules, allowing them to exist independently and causing the ionic substance to dissolve in water. Conversely, non-polar solvents such as benzene (C_6H_6) and hexane (C_6H_{14}) are unable to dissolve ionic compounds due to the absence of the necessary electrostatic interactions.





Characteristics of Ionic Compounds



➤ **Conductors of electricity**

Solid ionic compounds are poor conductors of electricity because the ions are rigidly fixed in their positions within the crystal lattice, preventing the flow of electrical charge. However, in the molten state or when dissolved in water, the ionic bonds break, and the ions become free to move. As a result, molten ionic compounds or their aqueous solutions can conduct electricity when placed in an electrolytic cell.

➤ **Do not exhibit isomerism**

The ionic bond, formed by the electrostatic attraction between oppositely charged ions, is non-rigid and non-directional. Consequently, ionic compounds lack the structural flexibility required for exhibiting isomerism.

➤ **Ionic reactions are fast**

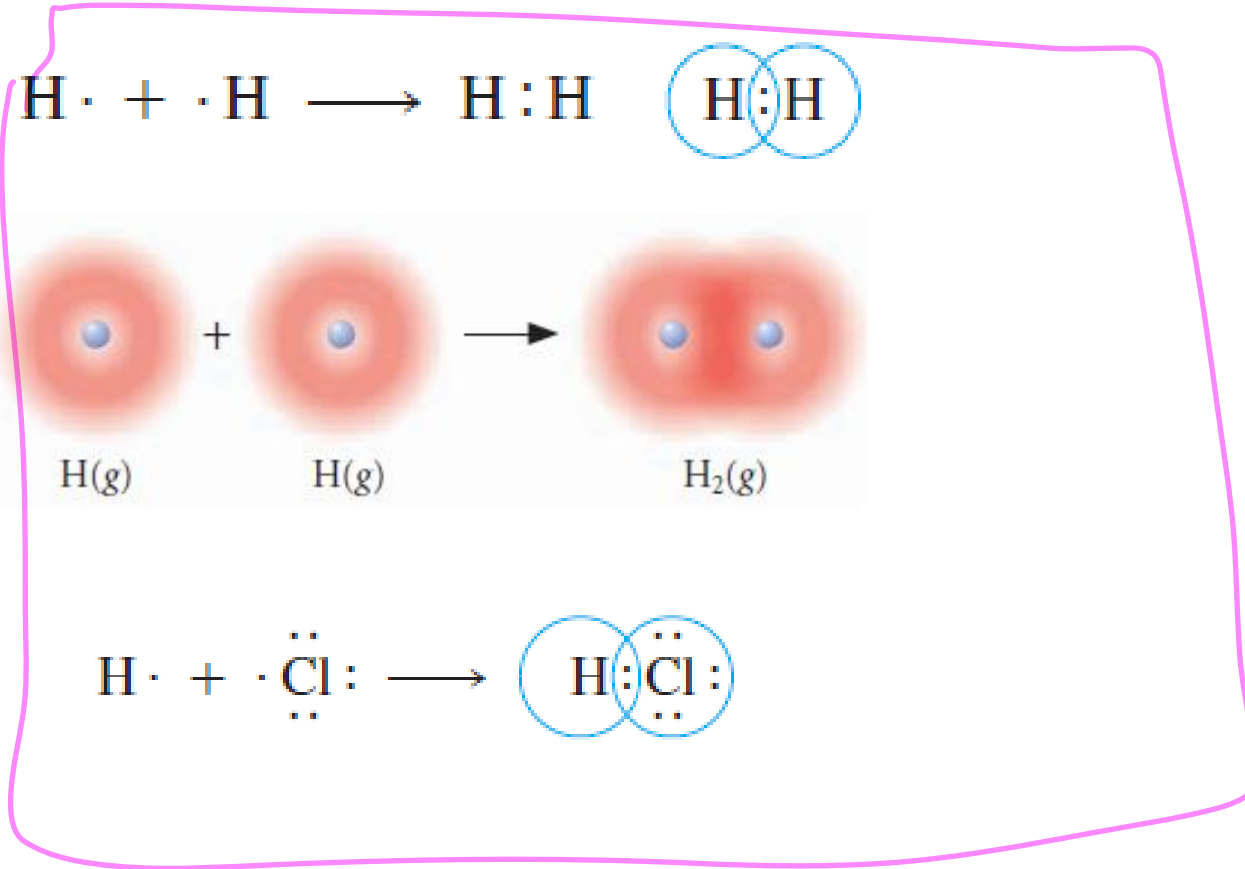
Ionic compounds give reactions between ions, and these are very fast.



Covalent Bond



Covalent bond is a chemical bond formed by the sharing of a pair of electrons between atoms.





Factors Favoring The Formation of Covalent Bonds



➤ Number of valence electrons

Each of the atom *A* and *B* should have 5, 6 or 7 valence electrons so that both achieve the stable octet by sharing 3, 2 or 1 electron-pair. H has one electron in the valence shell and attains duplet. The non-metals of groups VA, VIA and VIIA respectively satisfy this condition.

➤ Equal electronegativity

The atom *A* will not transfer electrons to *B* if both have equal (or nearly equal) electronegativity, and hence electron sharing will take place. This can be strictly possible only if both the atoms are of the same element.

➤ Equal sharing of electrons

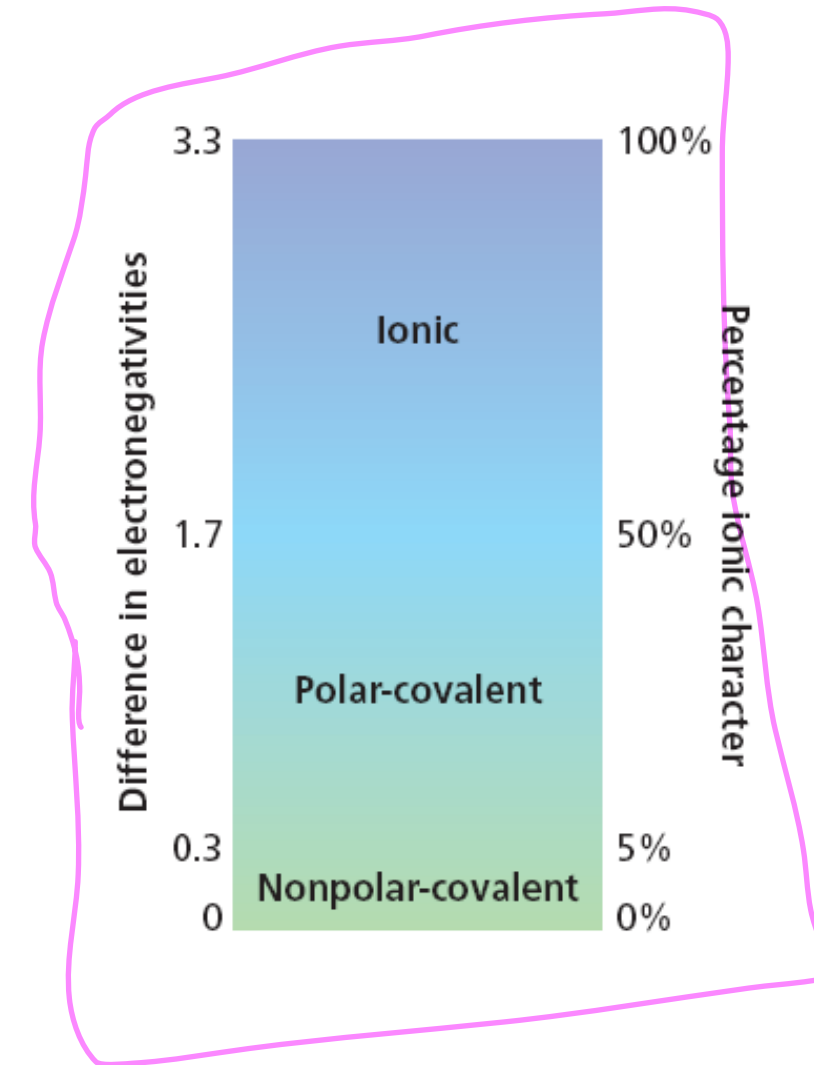
The atoms *A* and *B* should have equal (or nearly equal) electron affinity so that they attract the bonding electron pair equally. Thus equal sharing of electrons will form a nonpolar covalent bond. Of course, precisely equal sharing of electrons will not ordinarily occur except when atoms *A* and *B* are atoms of the same element, for no two elements have exactly the same electron affinity.



Factors Favoring The Formation of Covalent Bonds



- Extent of bonding between atoms of 2 elements being ionic or covalent depends on difference in electronegativity between two bonded atoms.
- Fluorine's EN = 4.0, Cesium's EN = 0.7
 $4.0 - 0.7 = 3.3$
- According to table, F-Cs is ionic
- The greater the difference, the more ionic the bond

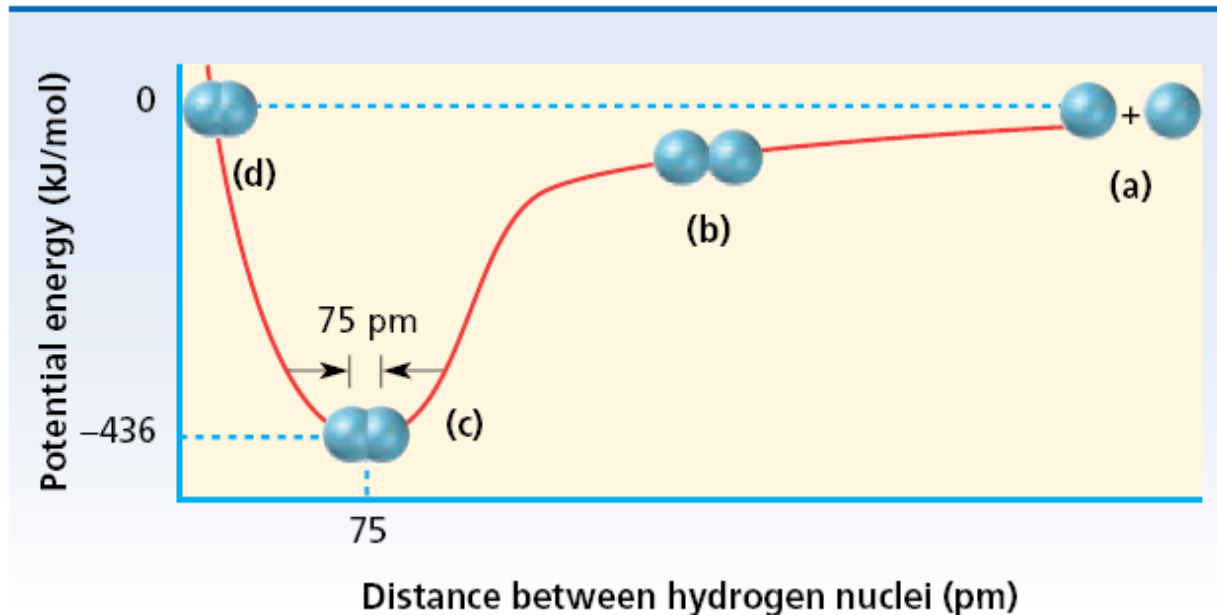




Energy Involved in Covalent Bonding



- Bonded atoms have lower potential energy than unbonded atoms
- At large distance atoms don't influence each other, and potential energy set at 0 at this stage.
- Consider formation of hydrogen molecules(H_2); Each H has (+) proton and nucleus surrounded by (-) electron
- As atoms near each other, charged particles start to interact
- Approaching nuclei and electrons are attracted to each other and decrease in total potential energy

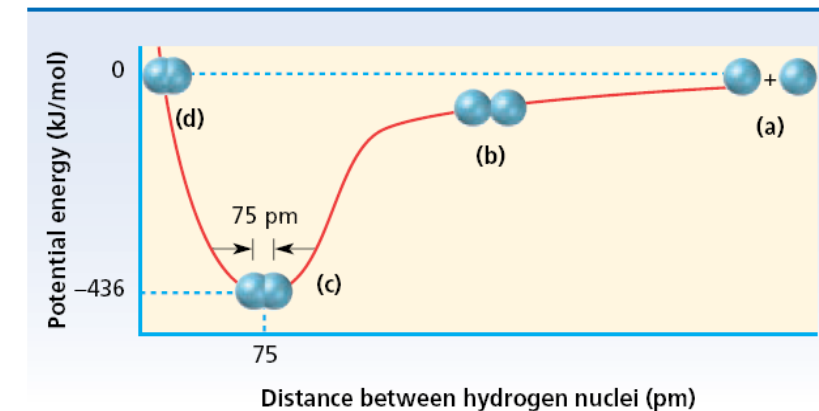
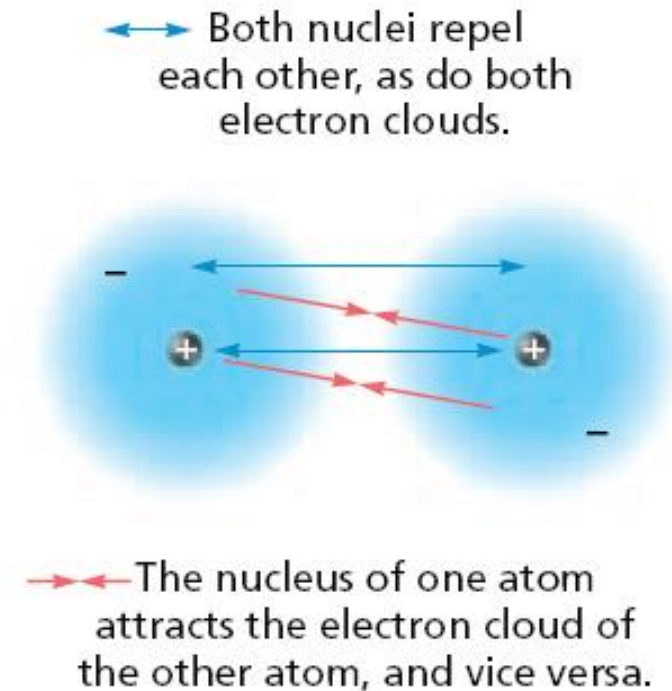




Energy Involved in Covalent Bonding



- The amount of attraction/repulsion depends on how close the atoms are to each other
- When atoms first “see” each other, electron-proton attraction stronger than e-e or p-p repulsions
- So atoms drawn to each other and potential energy lowered
- Attractive force dominates until a distance is reached where repulsion equals attraction
- As long as energy stays close to minimum, they stay covalently bonded (Valley of the curve)
- Closer the atoms get, potential energy rises sharply
- Repulsion becomes greater than attraction





Characteristics of Covalent Compounds

While the atoms in a covalent molecule are firmly held by the shared electron pair, the individual molecules are attracted to each other by weak van der Waals forces. Thus the molecules can be separated easily as not much energy is required to overcome the intermolecular attractions. This explains the general properties of covalent compounds.

➤ **Gases, liquids or solids at room temperature**

The covalent compounds are often gases, liquids or relatively soft solids under ordinary conditions. This is so because of the weak intermolecular forces between the molecules.

➤ **Low melting points and boiling points**

Covalent compounds have generally low melting points (or boiling points). The molecules are held together in the solid crystal lattice by weak forces. On application of heat, the molecules are readily pulled out and these then acquire kinetic energy for free movement as in a liquid.



Characteristics of Covalent Compounds

➤ Neither hard nor brittle

While the ionic compounds are hard and brittle, covalent compounds are neither hard nor brittle. There are weak forces holding the molecules in the solid crystal lattice. A molecular layer in the crystal easily slips relative to other adjacent layers and there are no 'forces of repulsion' like those in ionic compounds. Thus the crystals are easily broken and there is no sharp cleavage between the layers on application of external force.

➤ Soluble in organic solvents

In general, covalent compounds dissolve readily in nonpolar organic solvents (benzene, ether). The kinetic energy of the solvent molecules easily overcomes the weak intermolecular forces. Covalent compounds are insoluble in water. Some of them (alcohols, amines) dissolve in water due to hydrogen-bonding.

➤ Non-conductors of electricity

Since there are no (+) or (–) ions in covalent molecules, the covalent compounds in the molten or solution form are incapable of conducting electricity.



Characteristics of Covalent Compounds



➤ **Exhibit Isomerism**

Covalent bonds are rigid and directional, the atoms being held together by shared electron pair and not by electrical lines of force. This affords opportunity for various spatial arrangements and covalent compounds exhibit stereoisomerism.

➤ **Molecular reactions**

The covalent compounds give reactions where the molecule as a whole undergoes a change. Since there are no strong electrical forces to speed up the reaction between molecules, these reactions are slow.



Comparison of Ionic And Covalent Bonds

Ionic Bond	Covalent Bond
1. Formed by transfer of electrons from a metal to a non-metal atom. 2. Consists of electrostatic force between (+) and (–) ions. 3. Non-rigid and non-directional; cannot cause isomerism.	1. Formed by sharing of electrons between nonmetal atoms. 2. Consists of a shared pair of electrons between atoms. 3. Rigid and directional : causes stereoisomerism.
Properties of Compounds 1. Solids at room temperature. 2. High melting and boiling points. 3. Hard and brittle. 4. Soluble in water but insoluble in organic solvents. 5. Conductors of electricity 6. Undergo ionic reactions which are fast.	Properties of Compounds 1. Gases, liquids or soft solids. 2. Low melting and boiling points. 3. Soft, much readily broken 4. Insoluble in water but soluble in organic solvents. 5. Non-conductors of electricity. 6. Undergo molecular reactions which are slow.



Electronegativity and Covalent Bonds



Electronegativity is a measure of the ability of an atom in a molecule to draw bonding electrons to itself.

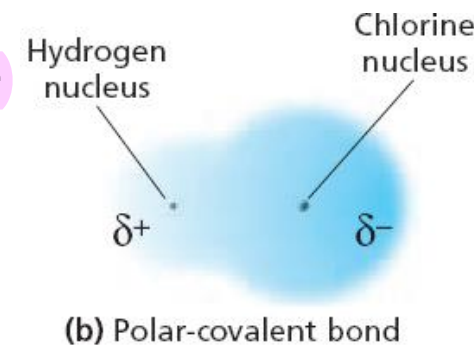
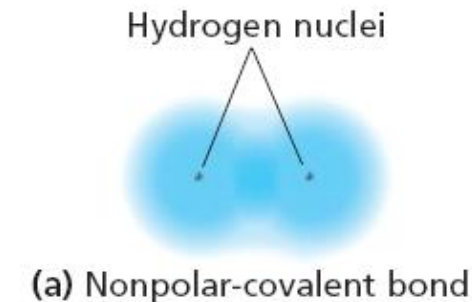
Nonpolar covalent Bond

- Nonpolar-covalent bond is a covalent bond in which the bonding electrons are shared equally by the bonded atoms, resulting in a balanced distribution of electrical charge
- 0-5% ionic character (0-0.3 EN difference) is nonpolar-covalent bond

Polar covalent bond

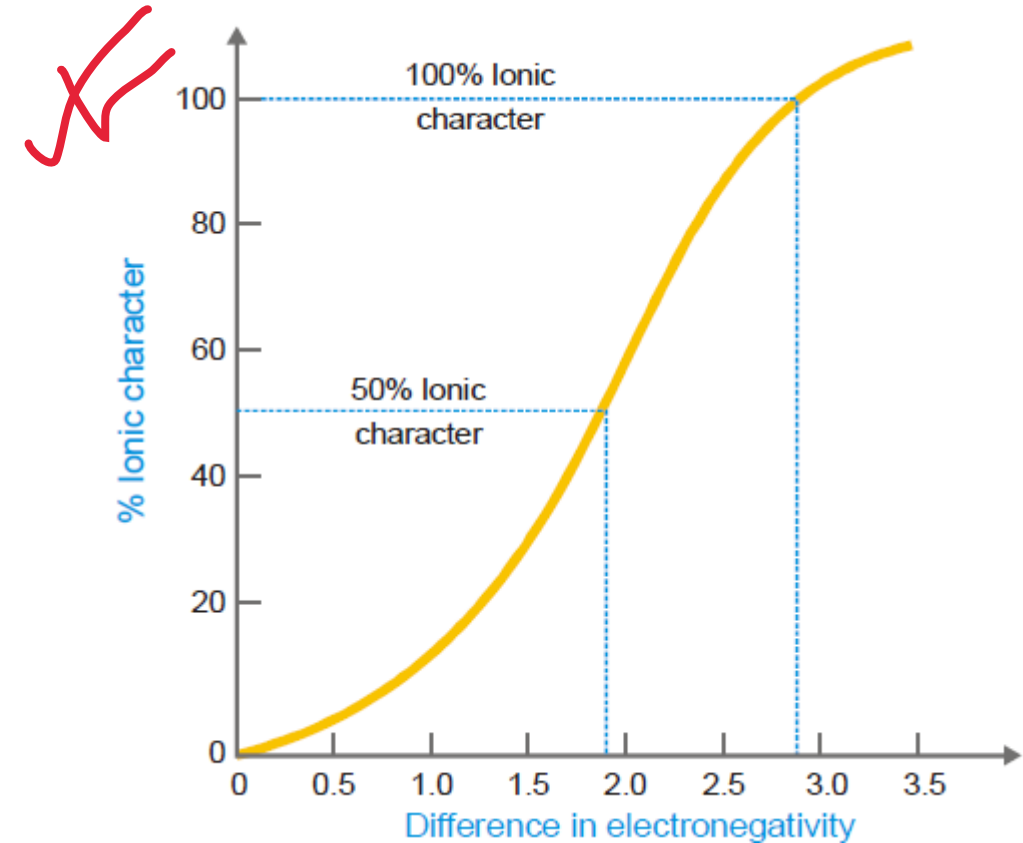
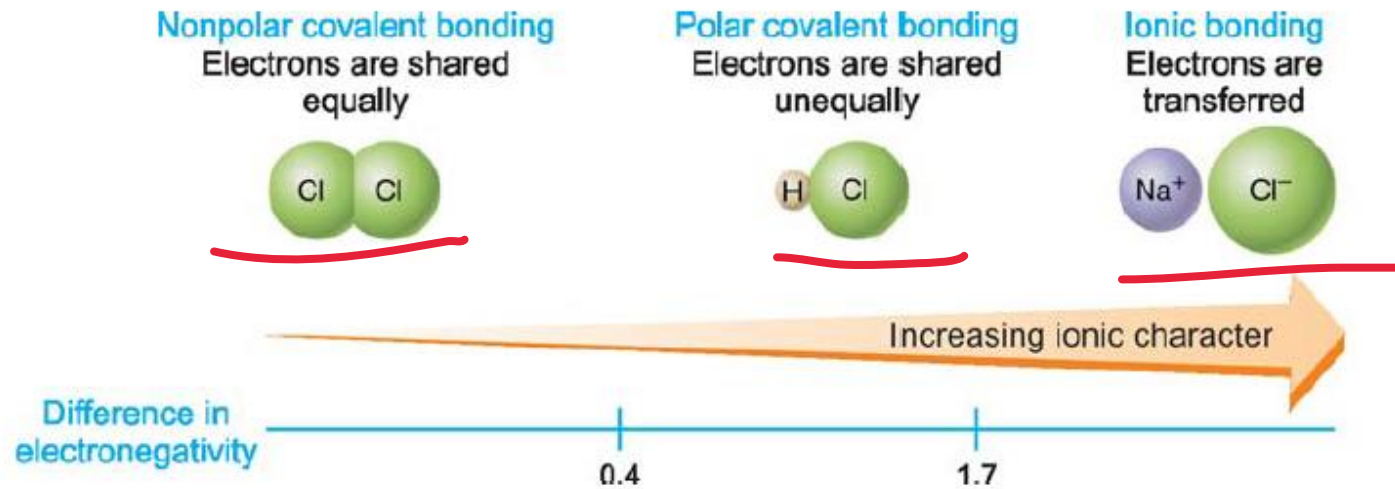
- ✓ ➤ Polar-covalent bond is a covalent bond in which the bonded atoms have an unequal attraction for the shared electrons
- Bonds that have significantly different ENs, electrons are more strongly attracted by more-EN atom
- These bonds are polar → *they have an uneven distribution of charge*

The greater the difference of electronegativity between two atoms, greater the polarity





Electronegativity and Covalent Bonds





Electronegativity and Covalent Bonds



The percentage ionic character of a bond can be calculated by using the equation

$$\% \text{ ionic character} = 16 [X_A - X_B] + 3.5 [X_A - X_B]^2$$

This equation was given by **Hannay and Smith**.

Calculate the percentage ionic character of C–Cl bond in CCl_4 if the electronegativities of C and Cl are 3.5 and 3.0 respectively.

$$\% \text{ ionic character} = 16 [X_A - X_B] + 3.5 [X_A - X_B]^2$$

Give $X_A = 3.5$ and $X_B = 3.0$

$$\begin{aligned} \therefore \% \text{a ionic character} &= 16 (3.5 - 3.0) + 3.5 (3.5 - 3.0)^2 \\ &= 8.0 + 0.875 \\ &= 8.875\% \end{aligned}$$



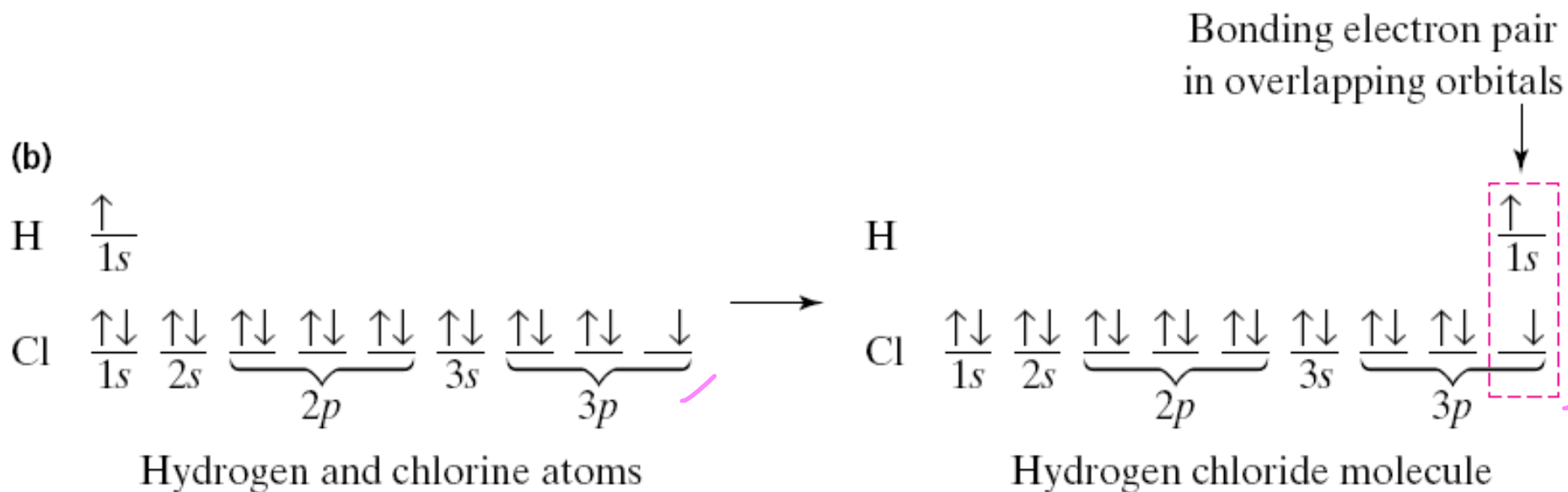
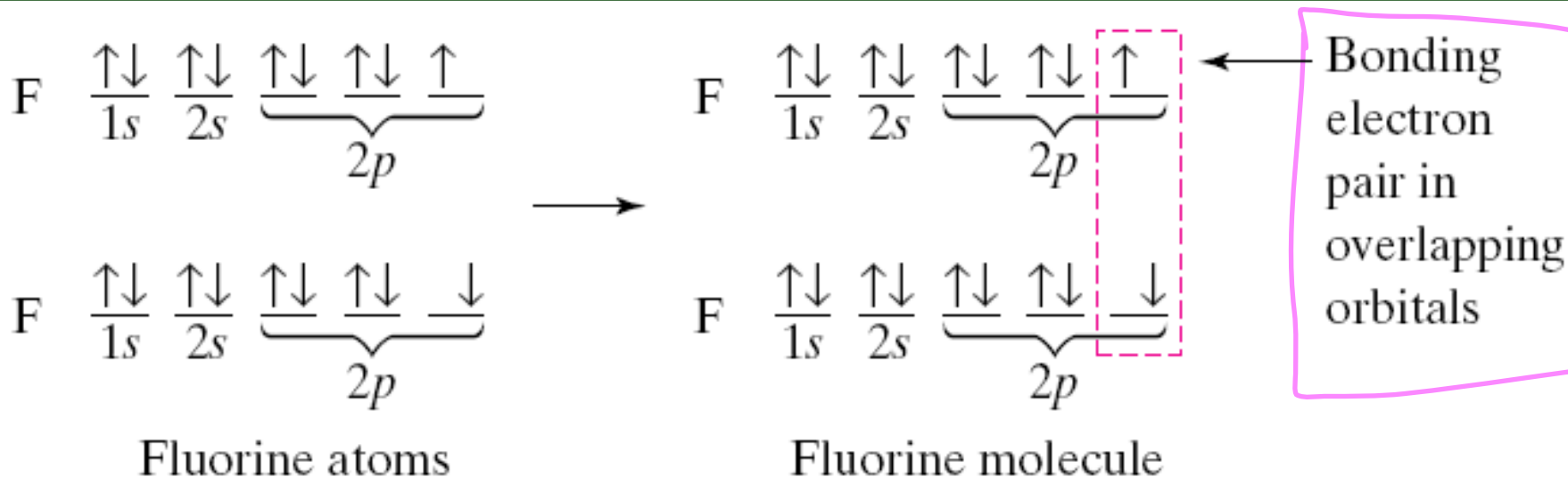
Octet Rule



- Noble-gas atoms have minimum energy existing on their own because of electronic configurations
- Outer orbitals completely filled
- Other atoms fill orbitals by transferring or sharing electrons
- Bond formation follows octet rule → chemical compounds tend to form so that each atom, by gaining, losing, or sharing electrons, has an octet of electrons in its highest occupied energy level



Octet Rule





Limitations of Octet Rule



- Most main-group elements form covalent bonds according to octet rule
- It is clear that octet rule is based upon the chemical inertness of noble gases. However, some noble gases (Xe, Kr) also combine with oxygen and fluorine to form a number of compounds like XeF_2 , KrF_2 , XeOF_2 etc. This theory does not account for the shape of molecules.
- Incomplete octet of the central atom e.g. H-H only 2 electrons
- Boron, B, has 3 valence electrons ($[\text{He}]2s^22p^1$) and boron tends to form bonds where it is surrounded by 6 e⁻ (e⁻ pairs)
- The Expanded Octet: Elements in and beyond the third period of the periodic table have, apart from 3s and 3p orbitals, 3d orbitals also available for bonding. In a number of compounds of these elements there are more than eight valence electrons around the central atom. This is termed as expanded octet. Examples- PF_5 , SF_6 , H_2SO_4